

BACHELOR OF TECHNOLOGY (CBCS) (2021-COURSE)
B. Tech. Sem - V INFORMATION TECHNOLOGY : SUMMER : 2024
SUBJECT: ITC-III: ADVANCED DATABASE SYSTEM

Day : Thursday
Date : 16/05/2024

S-25758-2024

Time : 10:00 AM-01:00 PM
Max. Marks : 60

N.B.

- 1) All questions are **COMPULSORY**.
- 2) Figures to the right indicate **FULL** marks.
- 3) Draw neat and labelled diagram **WHEREVER** necessary.
- 4) Assume suitable data if necessary.

Q.1 Elaborate distributed data processing. Enlist its advantages and disadvantages. [10]

OR

Q.1 Describe in detail the various transparencies in distributed database system. [10]

Q.2 Describe with suitable example distributed query optimization algorithm. [10]

OR

Q.2 Explain distributed query processing methodology along with translation of global queries into fragment queries. [10]

Q.3 Describe with reference to an example parallel query processing. [10]

OR

Q.3 Enlist and describe types of parallelism in distributed database system with an example. [10]

Q.4 Explain transaction management with ACID properties, deadlock detection and management and their algorithms. [10]

OR

Q.4 Explain concurrency control in centralized database system vs concurrency control in DDBS. [10]

Q.5 Describe following terms with an example [10]

- a) Decision support system
- b) Data analysis and OLAP
- c) Data mining and warehousing

OR

Q.5 Describe following terms with an example. [10]

- a) Benchmarking with different types
- b) Performance measurement and tuning.

Q.6 Explain features, types and query mechanism tools of NoSQL databases. [10]

OR

Q.6 Explain MongoDB Architecture, features and its modeling in detailed manner. [10]

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